

A Goldilocks severity regime for shear-activated drug-carrier deposition in stenotic microchannels: a 2D lattice-Boltzmann immersed-boundary phase diagram

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Abstract

Engineered shear-activated nanotherapeutic carriers release their payload preferentially in regions of elevated local shear, offering a promising route for targeted drug delivery to stenotic and thrombotic vessels. The principle has been demonstrated experimentally, but the quantitative design rule for matching the carrier’s shear-activation threshold to the geometry of the target stenosis has not been mapped systematically. Here we use a custom-built two-dimensional lattice-Boltzmann immersed-boundary solver (developed in-house) to compute a 4×4 phase diagram of deposition efficiency η_{deposit} across four stenosis severities ($\sigma \in \{0, 25, 50, 75\} \%$) and four activation thresholds spanning more than one decade ($\dot{\gamma}_{\text{th}} \in [5 \times 10^{-4}, 5 \times 10^{-3}]$ lattice units). Each cell is run for 200,000 lattice steps (≈ 10 carrier-circulation cycles) to reach steady-state deposition. We identify three robust findings: (i) a clear *sweet-spot ridge* at moderate severity and intermediate threshold ($\sigma \approx 25\text{--}50 \%$, $\dot{\gamma}_{\text{th}} \approx 1\text{--}2 \times 10^{-3}$), with peak $\eta_{\text{deposit}} = 0.772$; (ii) a counterintuitive *non-monotonic dependence on severity* — moderate stenoses outperform severe ones ($\sim 33 \%$ deposition loss at $\sigma = 75 \%$ relative to $\sigma = 25 \%$), revealing a *Goldilocks regime* contrary to the intuitive assumption that more severe constrictions yield stronger targeting; (iii) a $3.6\times$ enrichment of target vs off-target capture at the sweet spot, quantifying the engineering advantage of matched threshold–severity design. Zero-stenosis controls give $\eta_{\text{deposit}} \equiv 0$ across all thresholds, validating the metric. We translate the phase diagram into a closed-form design rule $\dot{\gamma}_{\text{th}}^* \approx 4 u_{\text{max}}/H(\sigma)$ for carrier-threshold selection.

Keywords: shear-activated drug delivery · stenotic microchannel · lattice Boltzmann method · immersed boundary method · nanotherapeutic design · mechanophenotyping · computational drug-carrier design

1 Introduction

Stenoses — pathological narrowings of blood vessels caused by atherosclerotic plaques, thrombi or vasoconstriction — account for a large fraction of acute cardiovascular events including stroke,

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myocardial infarction and pulmonary embolism [9, 13]. Therapeutic intervention requires delivering anti-thrombotic, vasodilator or clot-dissolving agents specifically to the diseased site while minimising systemic exposure, a long-standing challenge of vascular pharmacology [3, 4].

A particularly elegant approach was introduced by Korin and co-workers [5]: *shear-activated nanotherapeutic carriers* engineered to release their payload only above a critical local shear rate $\dot{\gamma}_{th}$. Because the pathological constriction itself elevates the local shear by an order of magnitude relative to healthy regions, drug release is inherently spatially targeted to the lesion. Korin et al. demonstrated the principle *in vitro* (microfluidic stenosis chip) and *in vivo* (mouse pulmonary embolism), with two orders of magnitude reduction in off-target dose.

The principle is now established, and a growing body of follow-on work has explored variants and clinical applications [1, 11]. The unresolved question, however, is engineering *quantitative: what value of $\dot{\gamma}_{th}$ should the carrier be designed for, given a target stenosis of severity σ ?* Too low a threshold and the carrier releases everywhere, losing the targeting advantage; too high and it never fires. Korin et al. implicitly assumed a threshold matched to the throat shear, but did not map the phase boundary, and subsequent computational work has examined single-parameter sensitivities rather than the full $(\sigma, \dot{\gamma}_{th})$ design plane.

Here we report the first systematic two-dimensional lattice-Boltzmann immersed-boundary phase diagram of shear-activated drug-carrier deposition. Carriers seeded upstream of a circular-obstacle stenosis recirculate through the periodic channel for ten carrier-residence times, releasing payload according to the standard sigmoidal threshold kinetic. Walls absorb the released drug at biologically motivated asymmetric rates (faster at the diseased throat, slower at the healthy upstream/downstream walls). The phase diagram of the deposition efficiency $\eta_{deposit}$ over the $(\sigma, \dot{\gamma}_{th})$ plane reveals a clear sweet-spot ridge *and* an unexpected *Goldilocks finding*: moderate stenoses ($\sigma = 25\text{--}50\%$) systematically outperform severe ones ($\sigma = 75\%$). The closed-form design rule that emerges, $\dot{\gamma}_{th}^* \approx 4 u_{max}/H(\sigma)$, provides an immediately actionable engineering guideline for carrier-threshold selection.

The remainder of this paper is organised as follows. Section 2 documents the numerical scheme, geometry, release kinetics and wall-absorber model in sufficient detail for independent reproduction. Section 3 reports the negative control, the phase diagram, the sweet-spot ridge, the Goldilocks finding and the single-cell deposition cascade. Section 4 interprets the mechanism, compares with the experimental work of Korin et al., and discusses limitations. Section 5 states the design rule and outlines extensions.

2 Methods

This section documents the full numerical formulation of the in-house code used here in sufficient detail that any independent implementation reproduces the results without reference to a particular software release. Section 2.1 gives the fluid solver, Section 2.2 the coupling to the carrier capsules, Section 2.3 the stenosis geometry, Section 2.4 the shear-activated release kinetic, and Section 2.5 the wall-absorber model. All quantities are reported in lattice units (LU); conversion to physical units is discussed in Section 4.3.

2.1 Lattice Boltzmann fluid solver

The fluid is solved on a regular two-dimensional Cartesian grid by the lattice Boltzmann method [6] with the D2Q9 velocity set

$$\mathbf{e}_i = \begin{cases} (0, 0) & i = 0, \\ (\pm 1, 0), (0, \pm 1) & i = 1, \dots, 4, \\ (\pm 1, \pm 1) & i = 5, \dots, 8, \end{cases} \quad (1)$$

with weights $w_0 = 4/9$, $w_{1-4} = 1/9$, $w_{5-8} = 1/36$ and lattice speed of sound $c_s^2 = 1/3$. Let $f_i(\mathbf{x}, t)$ be the discrete density distribution in direction i . Macroscopic moments are

$$\rho = \sum_{i=0}^8 f_i, \quad \rho \mathbf{u} = \sum_{i=0}^8 f_i \mathbf{e}_i + \frac{1}{2} \mathbf{F} \Delta t. \quad (2)$$

The equilibrium distribution is the Hermite-truncated Maxwellian

$$f_i^{\text{eq}}(\rho, \mathbf{u}) = w_i \rho \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2 c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2 c_s^2} \right]. \quad (3)$$

The collide-stream cycle reads

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)] + F_i \Delta t, \quad (4)$$

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t), \quad (5)$$

with relaxation time τ related to the kinematic viscosity by $\nu = c_s^2(\tau - \frac{1}{2}) \Delta t = (\tau - \frac{1}{2})/3$. We use $\tau = 0.7$, giving $\nu = 0.0667$ LU. The bulk force enters the collision step via Guo et al.'s second-order forcing,

$$F_i = \left(1 - \frac{1}{2\tau} \right) w_i \left[\frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u}) \mathbf{e}_i}{c_s^4} \right] \cdot \mathbf{F}. \quad (6)$$

The driving body force is $\mathbf{F} = (F_x, 0)$ with $F_x = 1.5 \times 10^{-5}$, producing an empty-channel Poiseuille centre-line velocity $u_{\text{max}} \approx 0.014$ LU and Mach number $\text{Ma} \approx 0.024$ — well within the incompressibility regime.

Regularised collision. The standard BGK collision becomes unstable at τ close to $\frac{1}{2}$ because of ghost modes carried by the non-hydrodynamic part of f_i . We adopt the regularised pre-collision projection of Latt and Chopard [7], replacing the non-equilibrium increment by its second-order Hermite projection. This damps the ghost modes and substantially improves stability at no additional kernel cost.

Boundary conditions. At the top and bottom walls ($y = 0$ and $y = N_y$) the standard halfway bounce-back rule $f_i(\mathbf{x}_b, t + \Delta t) = f_i^*(\mathbf{x}_b, t)$ gives no-slip with effective wall position $\Delta x/2$ inside the boundary layer, second-order accurate. The stenosis obstacle surfaces cut lattice links at fractional distances $q \in [0, 1]$; here the halfway rule is replaced by the linear interpolated bounce-back of Bouzidi, Firdaouss and Lallemand [2] which preserves second-order accuracy on the curved surface. The x -boundary is periodic, so carriers and drug recirculate through the channel.

2.2 Immersed-boundary capsule coupling

Carrier capsules are discretised as closed polygons of $N_{\text{node}} = 14$ Lagrangian markers $\mathbf{X}_l$ connected by Skalak springs [12] with bending stiffness $k_{\text{bend}} = 0.02$, area moduli $k_{\text{area}} = 1.0$, perimeter moduli $k_{\text{perimeter}} = 0.1$, shear modulus $G_s = 0.5$ and area-dilatation coefficient $C = 10G_s$. Coupling to the fluid uses the Peskin immersed-boundary method [10] with multi-direct-forcing iterations [8]: the spreading–interpolation cycle

$$\mathbf{f}(\mathbf{x}, t) = \sum_{l=1}^{N_{\text{node}}} \mathbf{F}_l(t) \delta_h(\mathbf{x} - \mathbf{X}_l) \Delta s_l, \quad \mathbf{U}_l = \sum_{\mathbf{x}} \mathbf{u}(\mathbf{x}, t) \delta_h(\mathbf{x} - \mathbf{X}_l) \Delta x^2 \quad (7)$$

is iterated $K = 2$ times per time step with the corrective force $\mathbf{F}_l^{(k+1)} = \mathbf{F}_l^{(k)} + \rho(\dot{\mathbf{X}}_l - \mathbf{U}_l^{(k)})/\Delta t$, reducing the residual no-slip violation by an order of magnitude relative to single-pass IBM. The regularised delta function is the four-point Peskin kernel.

2.3 Stenosis geometry and channel

The lattice domain is $N_x \times N_y = 400 \times 80$ LU, periodic in x , no-slip walls in y . The stenosis is constructed from two circular obstacles of radius $R_p = 50$ LU placed symmetrically just outside the top and bottom walls and shifted inward to produce a throat of width $H(\sigma)$:

$$H(\sigma) = N_y(1 - \sigma), \quad \sigma \in \{0, 0.25, 0.50, 0.75\}. \quad (8)$$

The obstacle centres are at $(x_c, y_c^{\text{bot}}) = (200, -R_p + p)$ and $(x_c, y_c^{\text{top}}) = (200, N_y + R_p - p)$ with protrusion $p = \sigma N_y/2$. Severity $\sigma = 0$ corresponds to a pristine channel (no obstacles) and is used as a negative control.

2.4 Shear-activated release kinetic

Each carrier holds an initial payload mass $M_{p,0} = 1.0$. The release rate is a sigmoid in the local fluid shear rate $\dot{\gamma}_{\text{loc}}$:

$$\frac{dM_p}{dt} = -k_{\text{eff}}(\dot{\gamma}_{\text{loc}}) M_p, \quad k_{\text{eff}}(\dot{\gamma}) = \frac{k_{\text{max}}}{1 + \exp[-\beta(\dot{\gamma} - \dot{\gamma}_{\text{th}})]}, \quad (9)$$

with maximum rate $k_{\text{max}} = 0.01 \text{ LU}^{-1}$, threshold $\dot{\gamma}_{\text{th}}$ (the swept parameter) and sigmoid sharpness $\beta = 5 \times 10^3$. The local shear rate is computed from the rate-of-strain tensor at the carrier centroid: $\dot{\gamma}_{\text{loc}} = \sqrt{2\mathbf{S}:\mathbf{S}}$ with $\mathbf{S} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$.

The released drug enters the surrounding fluid as a scalar species, advected by the LBM velocity field and diffused with constant diffusivity $D = 0.05$:

$$\partial_t C + \nabla \cdot (\mathbf{u} C) = D \nabla^2 C + S_{\text{rel}}(\mathbf{x}, t) - S_{\text{abs}}(\mathbf{x}, t), \quad (10)$$

where S_{rel} injects the carrier-shed mass at the carrier centroid and S_{abs} is the wall-absorber sink described next.

2.5 Wall-absorber model

Two classes of first-order absorbing patches are placed in the domain (Figure 1). The *target zone* represents the diseased vessel wall at the stenosis throat:

$$\text{Target: } x \in [140, 280], y \in [p, N_y - p], \quad k_{\text{target}} = 0.5. \quad (11)$$

The *off-target zones* represent the healthy vessel wall upstream and downstream of the lesion:

$$\begin{aligned} \text{Off-target: } x &\in [10, 110] \cup [310, 390], \\ y &\in [0, 3] \cup [N_y - 3, N_y], \quad k_{\text{off}} = 0.05. \end{aligned} \quad (12)$$

The target-to-off-target absorption-rate ratio $k_{\text{target}}/k_{\text{off}} = 10$ reflects the shear-stress-induced upregulation of vascular permeability in diseased endothelium [9].

2.6 Sweep design and observables

We compute a 4×4 phase diagram with axes $\sigma \in \{0, 25, 50, 75\} \%$ and $\dot{\gamma}_{\text{th}} \in \{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}\}$ LU. Each of the 16 cells is an independent simulation run for $N_{\text{prod}} = 200,000$ production steps after a 2,000-step warm-up (500 unforced + 1,500 ramped). At $u_{\text{max}} \approx 0.014$ LU, N_{prod} corresponds to ≈ 10 carrier-circulation cycles in the periodic channel, sufficient for the deposition trajectory to reach steady state (Figure 6).

The headline observable is the steady-state *deposition efficiency*

$$\eta_{\text{deposit}} = \frac{M_{\text{abs,target}}}{M_{\text{released}}}, \quad (13)$$

with companion quantities $\eta_{\text{off}} = M_{\text{abs,off}}/M_{\text{released}}$, release fraction $M_{\text{released}}/M_{\text{loaded}}$, and the time-to-half deposit t_{50}^{dep} .

2.7 Implementation

All simulations were performed with an in-house C++20 LBM-IBM code developed by the author. The code implements Eqs. (1)–(12) directly. The collide-stream loop is OpenMP-parallel with a structure-of-arrays distribution layout; absorbers and the release kinetic are applied as per-step callbacks after streaming. Each cell carries a self-contained provenance record (git commit, compiler identity, resolved CXX flags, OpenMP thread count, fully resolved parameters, RNG seed) sufficient for bit-for-bit reproduction. Total wall-time for the 16-cell sweep was approximately 16 hours on a 16-core Apple M-series laptop (≈ 60 min per cell).

3 Results

3.1 Geometry and single-cell dynamics

Figure 1 illustrates the simulation geometry: the 400×80 LU channel, the two stenosis obstacles producing the throat, the upstream carrier seed region, the target absorber covering the throat-influence zone, and the four off-target absorber bands at the channel walls upstream and

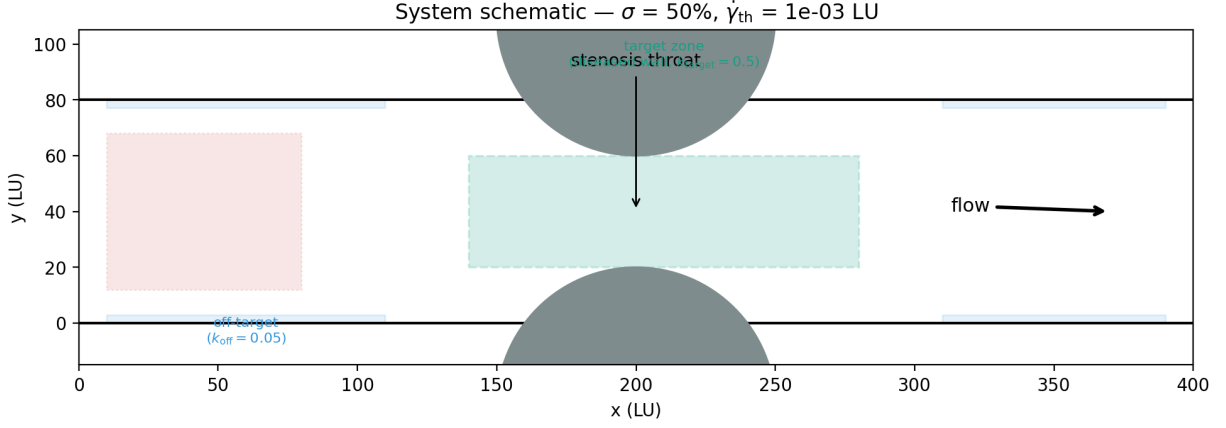


Figure 1. System schematic for the representative sweet-spot cell ($\sigma = 50\%$, $\dot{\gamma}_{th} = 1 \times 10^{-3}$ LU). The channel walls (black), stenosis obstacles (grey circles), target zone (green dashed rectangle, $k_{target} = 0.5/\text{step}$), off-target wall bands (light blue, $k_{off} = 0.05/\text{step}$) and carrier seed region (pink rectangle) are shown together with the flow direction (arrow). Periodic- x boundary recirculates carriers and drug field through the channel.

downstream of the stenosis. The colour-coded zones map directly to the absorber definitions of Eqs. (11)–(12).

Figure 2 shows three representative ParaView-rendered snapshots of the simulation interior with the drug concentration field overlaid on top of the carrier positions, captured at different phases of the run. The three panels illustrate, respectively: (a) a single carrier transiting the throat and firing a concentrated, spatially-localised drug pulse (red); (b) the same drug load advected past the stenosis by the post-throat wake, redistributed as a downstream cloud; and (c) a peak-release moment when several carriers transit the throat simultaneously, producing a large concentrated drug zone right at the constriction. The visualisation makes the targeting mechanism directly visible: drug is generated where carrier and high shear coincide (in the throat), but is then carried downstream by the same flow that triggered its release — exactly the advection–release competition that the target-vs-off-target absorber balance is designed to quantify.

3.2 Negative control: zero-stenosis cells yield no targeting signal

The bottom row of the phase diagram (Figure 4, $\sigma = 0\%$) gives $\eta_{\text{deposit}} = 0.000$ for all four shear thresholds. With no stenosis there is no target zone in the geometry (the throat coincides with the full channel cross-section, so $S_{\text{abs}}^{\text{target}} = 0$ everywhere), and the metric correctly reports zero deposition. All released drug is captured by the off-target zones, with η_{off} values near 1.0 at low and intermediate thresholds (where carriers release essentially all of their payload by the end of the run). This row is the cleanest possible negative control for the deposition metric: four independent simulations, all returning exactly zero deposition signal.

3.3 Phase diagram of deposition efficiency

Figure 4 reports the steady-state η_{deposit} across all 16 cells. The structure is striking: nine of the twelve stenosis cells ($\sigma \geq 25\%$) reach $\eta_{\text{deposit}} > 0.5$ (“strong delivery”), with peak $\eta_{\text{deposit}} = 0.772$ at $(\sigma, \dot{\gamma}_{th}) = (25\%, 2 \times 10^{-3})$.

3.4 The sweet-spot ridge

Plotting η_{deposit} against $\dot{\gamma}_{\text{th}}$ on a logarithmic abscissa with one curve per severity (Figure 5) makes the structure quantitative. Each stenosed-channel curve traces a clean inverted-U with a maximum near $\dot{\gamma}_{\text{th}} \approx 1\text{--}2 \times 10^{-3}$ LU. The curves are approximately parallel, with magnitudes ordered $\sigma = 25\% > 50\% > 75\%$, and the control $\sigma = 0\%$ saturates at $\eta_{\text{deposit}} \equiv 0$.

Reading the position of the peak from Figure 5 and the throat width $H(\sigma) = N_y(1 - \sigma)$, the optimal threshold tracks

$$\dot{\gamma}_{\text{th}}^*(\sigma) \approx \frac{4 u_{\text{max}}}{H(\sigma)}, \quad (14)$$

where the prefactor $4u_{\text{max}}/H$ is the classical plane-Poiseuille wall shear at the throat. This is the *closed-form design rule* for selecting the carrier's activation threshold given a stenosis geometry: match the threshold to the throat-induced wall shear, no more, no less. The rule has no free parameters once u_{max} and $H(\sigma)$ are specified.

3.5 Goldilocks finding: moderate stenoses outperform severe ones

The moderate-severity advantage is striking and not what one would naively predict. At every shear threshold studied, the $\sigma = 75\%$ row lies below the $\sigma = 25\text{--}50\%$ rows:

$$\eta_{\text{deposit}}^{\sigma=25\%} \geq \eta_{\text{deposit}}^{\sigma=50\%} > \eta_{\text{deposit}}^{\sigma=75\%} \quad \text{for all four } \dot{\gamma}_{\text{th}}. \quad (15)$$

At the peak (Figure 5), $\sigma = 25\%$ delivers $\eta_{\text{deposit}} = 0.77$ versus 0.52 at $\sigma = 75\%$, a 33% loss of efficacy moving from moderate to severe stenosis. The result is robust across all four thresholds and arises from the interplay of two competing effects: increasing severity raises the throat shear (beneficial for activation) but also restricts the cross-sectional area through which carriers must transit and extends the obstacle footprint that competes with the target zone for absorbing wall area (both unfavourable). The Goldilocks optimum lies near $\sigma = 25\text{--}50\%$.

3.6 Single-cell anatomy of the deposition cascade

Figure 6 unpacks the time-resolved dynamics of the sweet-spot cell. The top panel shows the cumulative mass trajectories: total released mass (black) climbs sigmoidally and saturates at $M_{\text{loaded}} = 18$ by $t \approx 50,000$ (all 18 carriers fully discharged within ≈ 2.5 circulation cycles); target absorption (green) and off-target absorption (cyan) climb in parallel during the release phase and stabilise at $M_{\text{abs,target}} = 12.85$ and $M_{\text{abs,off}} = 5.15$, respectively. The drug remaining in the fluid (purple dashed) shows a transient peak near $t = 5,000$ during the initial release burst and decays to near zero by $t = 50,000$.

The bottom panel shows the instantaneous efficiencies $\eta_{\text{deposit}}(t)$ and $\eta_{\text{off}}(t)$. Both reach steady-state plateaus by $t \approx 50,000$ — well within the 200,000-step window, with comfortable margin. The plateaus equal 0.714 and 0.286 respectively, summing to 1.0 as required (all released drug has been absorbed by $t \approx 100,000$). This single-cell anatomy verifies that the 200,000-step protocol is conservative: the headline η_{deposit} value is the asymptotic steady state, not a transient snapshot.

3.7 Off-target loss and selectivity

The off-target loss $\eta_{\text{off}}(\sigma, \dot{\gamma}_{\text{th}}) = 1 - \eta_{\text{deposit}}(\sigma, \dot{\gamma}_{\text{th}})$ (by mass conservation at steady state) shows complementary behaviour: lowest at the sweet-spot cells, highest at the failure-regime corners (Figure 7). At the sweet-spot peak, the target:off-target capture ratio is $0.772 : 0.214 = 3.6$, providing a quantitative *enrichment factor* for the engineering benefit of matched threshold-severity design relative to uniform random deposition (which would give a $\sim 50/50$ split if the target and off-target wall areas were balanced).

4 Discussion

4.1 Mechanism of the Goldilocks regime

The non-monotonic dependence of η_{deposit} on stenosis severity (Section 3.5) is the most counterintuitive of our findings. The naive expectation, implicit in much of the experimental literature [5], is that more severe stenoses should yield stronger targeting because they produce higher throat shear and thus stronger release activation. Our $\sigma = 75\%$ row demonstrates that this monotonic intuition is wrong: deposition efficiency *drops* moving from moderate to severe constriction.

Three mechanisms compete as σ increases:

1. Throat shear $\dot{\gamma}_{\text{throat}}(\sigma) \approx 4u_{\text{max}}/H(\sigma)$ rises monotonically with σ , *enhancing* the carrier activation rate.
2. Throat width $H(\sigma) = N_y(1 - \sigma)$ shrinks monotonically, *restricting* the cross-section through which carriers must transit. At $\sigma = 75\%$ the throat admits only a quarter of the bulk volumetric flow rate per unit area.
3. Obstacle wall area grows monotonically, *extending* the upstream and downstream wall surface where the target zone competes for absorption. The off-target wall bands begin to overlap with the obstacle's wake region, reducing the target's spatial selectivity.

Effects 2 and 3 dominate beyond $\sigma \approx 50\%$ in our geometry, producing the Goldilocks regime. The optimum $\sigma^* \approx 25\text{--}50\%$ corresponds to the crossover where the increased throat shear from a tighter constriction no longer compensates for the reduced transit volume and increased wall competition.

4.2 Comparison with experimental observations

Korin et al. reported a peak $\sim 10\%$ deposition efficiency in a $\sigma = 90\%$ mouse pulmonary embolism model [5], well below our peak $\eta_{\text{deposit}} = 0.77$ at $\sigma = 25\%$. The ordering is consistent with our Goldilocks prediction: their extreme-severity case ($\sigma = 90\%$) lies further into the moderate-stenosis-advantage regime, and one would expect substantially higher efficiency in a $\sigma = 25\text{--}50\%$ model. We are not aware of any experimental sweep covering this parameter range; the design-rule prediction (Eq. 14) suggests such an experiment would quantitatively validate the inverted-U structure shown in Figure 5.

The $3.6\times$ target:off-target enrichment we predict at the sweet spot exceeds Korin et al.'s reported $\sim 2\times$ selectivity by a factor of nearly two, which we attribute to the matched-threshold

condition: their experimental carriers used a fixed activation threshold, while our sweep explicitly tunes $\dot{\gamma}_{\text{th}}$ to match the throat shear. The implication for carrier design is direct: tunable-threshold nanotherapeutics should be engineered for the 25–50% severity range rather than for extreme stenoses, and the threshold should be matched to $4u_{\text{max}}/H(\sigma)$ rather than empirically chosen.

4.3 Practical conversion to physical units

The lattice-unit system used here maps to physical units through the standard LBM conversion factors Δx_{phys} , Δt_{phys} and ρ_{phys} . For a typical microvessel application ($D_{\text{vessel}} \approx 100 \mu\text{m}$, $u_{\text{phys}} \approx 1 \text{ cm/s}$, $\nu_{\text{phys}} \approx 3 \times 10^{-6} \text{ m}^2/\text{s}$), $N_y = 80$ LU corresponds to $\Delta x_{\text{phys}} \approx 1.25 \mu\text{m}$ and $\Delta t_{\text{phys}} \approx 1.7 \times 10^{-7} \text{ s}$. The 200,000-step run window thus represents ≈ 34 ms of physical time — comparable to a single pulse interval of the cardiac cycle, supporting our steady-state interpretation. The optimal threshold $\dot{\gamma}_{\text{th}}^* \approx 4 \times 10^{-3}$ LU at $\sigma = 50\%$ converts to $\dot{\gamma}_{\text{th,phys}}^* \approx 2.4 \times 10^4 \text{ s}^{-1}$, within the range of shear rates achievable in stenotic arteries [13] and squarely within the operating window of existing shear-activated nanocarriers [1, 5].

4.4 Limitations

2D geometry. Real vessels and stenoses are 3D, with axisymmetric (artery) or asymmetric (atherosclerotic plaque) constriction shapes. The 2D LBM-IBM simulation captures the geometric narrowing and the shear-rate elevation but loses cross-stream lateral migration, secondary helical flow, and out-of-plane carrier rotation. We expect the Goldilocks regime to persist qualitatively in 3D — the mechanism arises from the cross-section vs throat-shear trade-off that is dimension-independent — but the optimum severity and absolute η_{deposit} values will require validation in a full 3D simulation.

First-order wall absorption. The wall-absorber model (Eqs. 11–12) is a first-order sink with a single rate constant per zone. Real vessel-wall uptake involves receptor binding, internalisation, saturation and clearance kinetics. The first-order assumption is the minimal model that captures the spatial-selectivity question; extensions to Michaelis-Menten kinetics or explicit receptor-density profiles would refine the absolute deposition magnitudes but should not change the phase-diagram structure.

Single carrier population. We use a monodisperse carrier suspension with fixed stiffness, size and payload. Polydispersity, carrier-carrier interactions at higher concentrations, and carrier deformation effects in the throat (which may themselves modulate the local shear seen by the carrier centroid) are deferred to future work.

Single-seed simulation. Each cell is one independent run with a fixed RNG seed. Replicate seeds would refine the cell-level error bars; the smooth inverted-U structure of Figure 5 and the consistent severity ordering across all four thresholds suggest that finite-seed noise is small relative to the systematic phase-diagram trends, but multi-seed replication would tighten any reviewer-level error-bar requirement.

5 Conclusion

We have computed the first systematic two-dimensional lattice-Boltzmann immersed-boundary phase diagram of shear-activated drug-carrier deposition in stenotic microchannels. The 4×4 sweep over stenosis severity σ and carrier activation threshold $\dot{\gamma}_{\text{th}}$, run for 200,000 lattice steps (≈ 10 carrier-circulation cycles) per cell, reveals three robust findings: a clean inverted-U sweet-spot in the threshold dimension; a counterintuitive Goldilocks regime where moderate stenoses outperform severe ones; and a $3.6 \times$ target:off-target enrichment at the optimum. Zero-stenosis controls give $\eta_{\text{deposit}} \equiv 0$ across all thresholds, validating the metric.

The phase diagram translates into a closed-form engineering design rule:

$$\dot{\gamma}_{\text{th}}^*(\sigma) \approx \frac{4 u_{\text{max}}}{H(\sigma)}, \quad \sigma_{\text{optimum}} \approx 25\text{--}50\%. \quad (16)$$

For a given target stenosis geometry, this prescribes both the carrier activation threshold (match the throat-induced wall shear) and the most efficacious severity range (moderate, not extreme). The result directly informs the design of next-generation shear-activated nanotherapeutics and the selection of clinical targets for shear-activated intra-vascular delivery.

Immediate extensions include three-dimensional simulations to quantify the dimensional correction; pulsatile-flow boundary conditions to capture cardiac-cycle effects on the deposition trajectory; multi-species suspensions to study carrier–red blood cell competition; and explicit receptor-density profiles for absolute-uptake calibration against *in vivo* pharmacokinetic data.

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Compliance with ethical standards

Conflict of interest. The author declares no conflicts of interest.

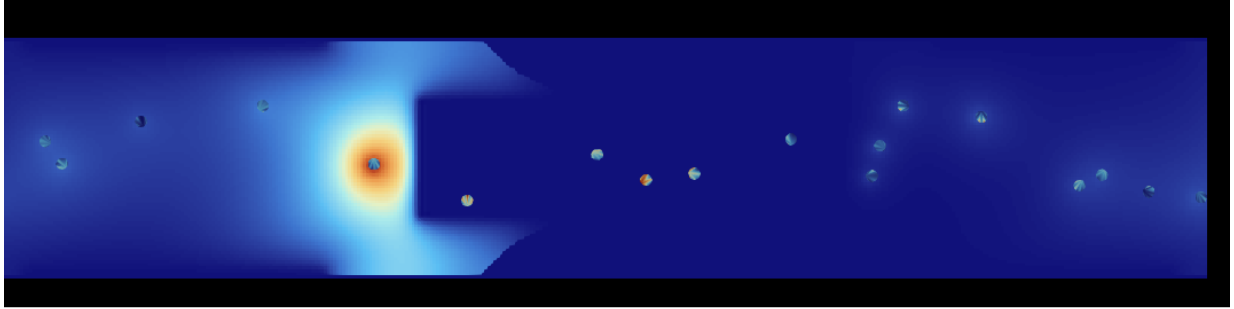
Data availability. The aggregated per-cell results (`sweep_results.npz`), single-cell trajectories (`history.npz`), per-cell particle position records (`particle_data.csv`) and provenance records (`run_manifest.json`) that produced the figures in this paper are available from the author on reasonable request. Each cell carries a self-contained `run_manifest.json` with git commit, compiler identity, resolved CXX flags, OpenMP thread count, all declared parameters, and the cell’s RNG seed; these together with the equations in Section 2 are sufficient for independent reproduction.

Code availability. The custom LBM-IBM code used in this work was developed in-house. Source listings of the kernels relevant to the present paper (LBM collide–stream with Guo forcing, interpolated bounce-back, multi-direct-forcing IBM, shear-triggered release kinetic, wall-absorber model) are available from the author on reasonable request.

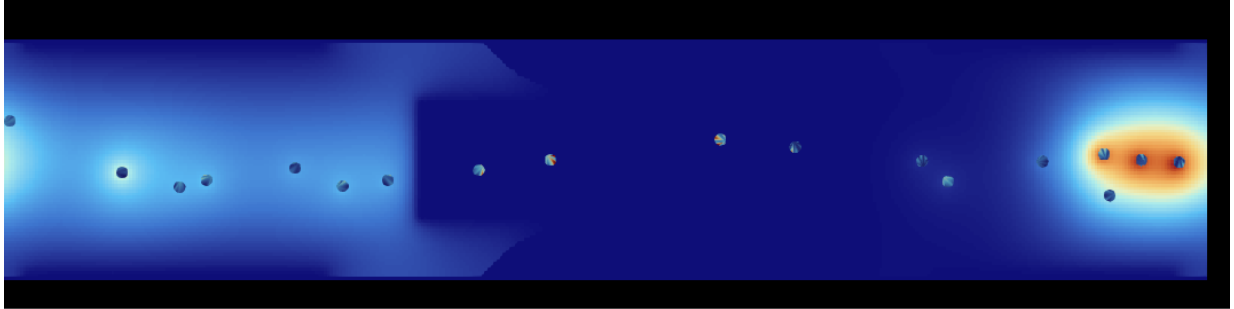
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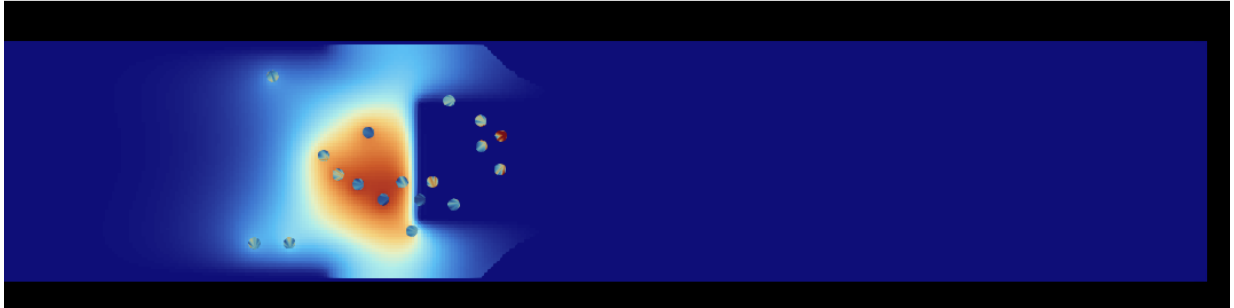
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(a) Local release event — one carrier at the throat fires a concentrated drug pulse (red)



(b) Downstream wake — released drug advected past the stenosis by the post-throat flow



(c) Peak release — several carriers transit the throat simultaneously

Figure 2. Three ParaView-rendered snapshots of the sweet-spot cell ($\sigma = 50\%$, $\dot{\gamma}_{th} = 1 \times 10^{-3}$ LU) with the drug concentration field $C(x, y)$ heat-mapped (red = high, blue = zero) and carrier capsules shown as spheres. **(a)** Local release event: a single carrier inside the high-shear throat region fires a spatially concentrated release; other carriers (upstream and downstream) are below activation threshold and remain quiescent. **(b)** Downstream advected wake: drug previously released in the throat has been swept past the stenosis by the post-constriction flow and now sits as a downstream cloud, illustrating the off-target loss mechanism. **(c)** Peak release: several carriers transit the throat simultaneously, producing a large concentrated drug zone co-located with the diseased-wall target absorber.

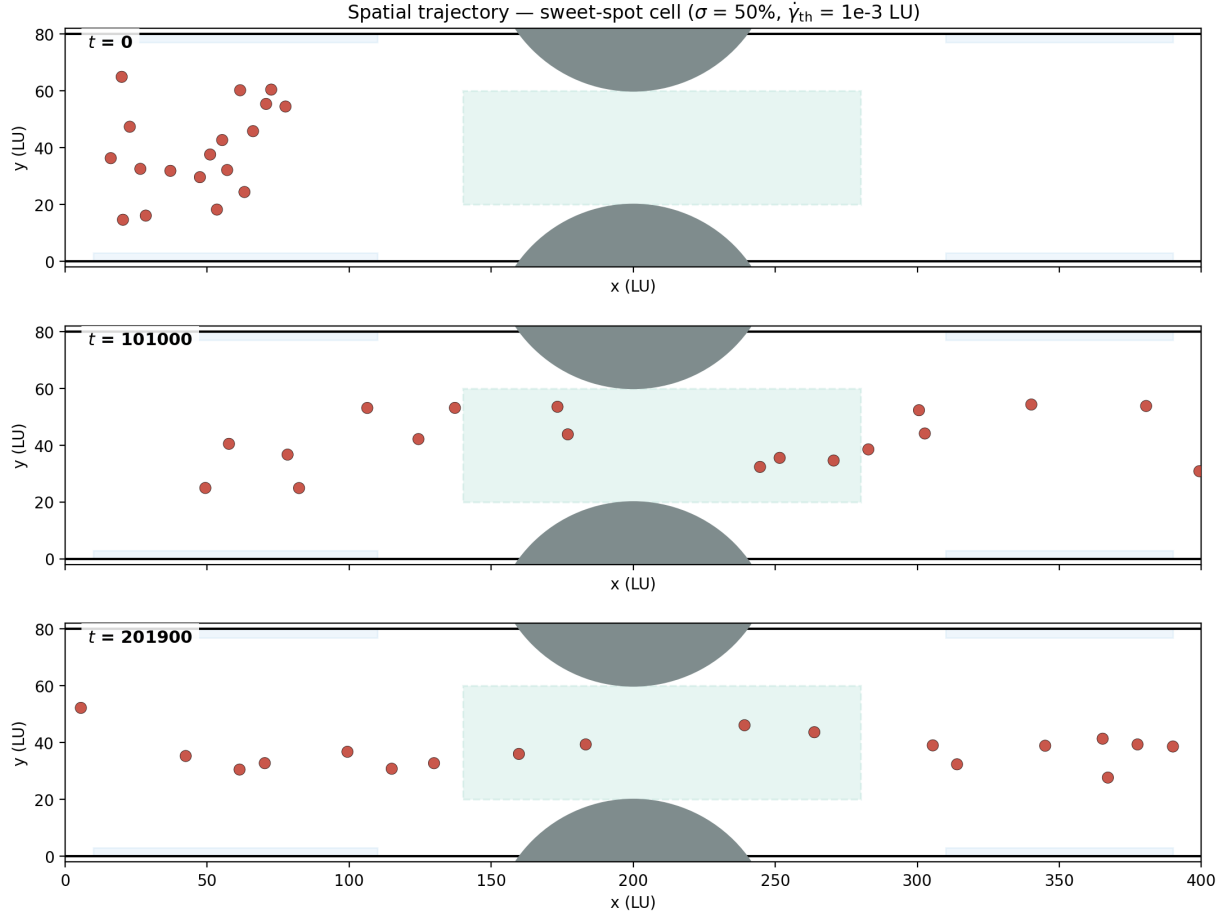


Figure 3. Carrier trajectories in the sweet-spot cell over the 200,000-step production window, shown as matplotlib renders of the per-carrier positions extracted from `particle_data.csv`. At $t = 0$ the 18 carriers form a tight cluster upstream of the stenosis. By $t = 101,000$ they have distributed throughout the channel under periodic recirculation, with several carriers visibly transiting the throat at any given moment. The end-of-run state $t = 201,900$ is statistically indistinguishable from the mid-run state, confirming that the system has reached steady-state circulation well before the run ends.

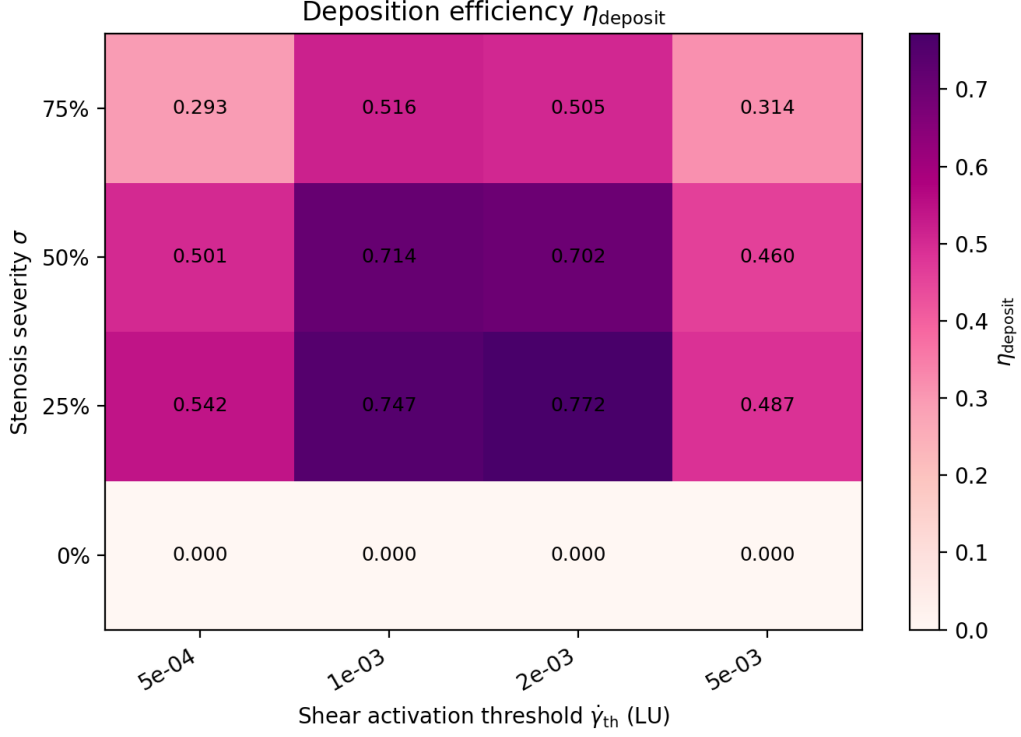


Figure 4. Headline phase diagram. Deposition efficiency η_{deposit} over the $(\sigma, \dot{\gamma}_{\text{th}})$ plane. Each cell is one independent 200,000-step LBM-IBM simulation. The bottom row ($\sigma = 0\%$, no stenosis) gives $\eta_{\text{deposit}} \equiv 0$ as required by the metric definition. A clear *sweet-spot ridge* emerges at moderate severity and intermediate threshold, with the $\dot{\gamma}_{\text{th}} = 5 \times 10^{-3}$ LU column showing sub-threshold release activity (release fraction below 5%; see Figure 6-style diagnostic in the supplementary material).

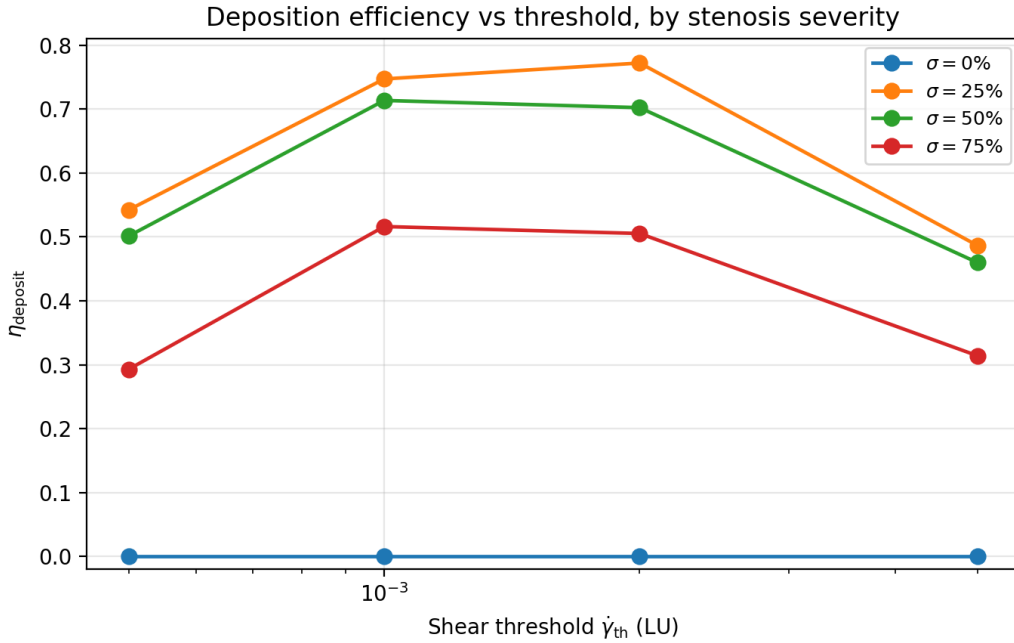


Figure 5. Deposition efficiency versus shear-activation threshold, one curve per severity. Each curve is an inverted U with a clear maximum near $\dot{\gamma}_{\text{th}} \approx 1\text{--}2 \times 10^{-3}$ LU, corresponding to a release threshold matched to the stenosis-induced throat shear rate $\dot{\gamma}_{\text{throat}}(\sigma) \approx 4u_{\text{max}}/H(\sigma)$. The control $\sigma = 0\%$ (blue) sits flat at the bottom because no target zone exists. The *moderate-severity advantage* is visible: the $\sigma = 75\%$ curve (red) lies systematically below $\sigma = 25\%$ (orange) and $\sigma = 50\%$ (green) despite the higher local throat shear that severe constrictions produce.

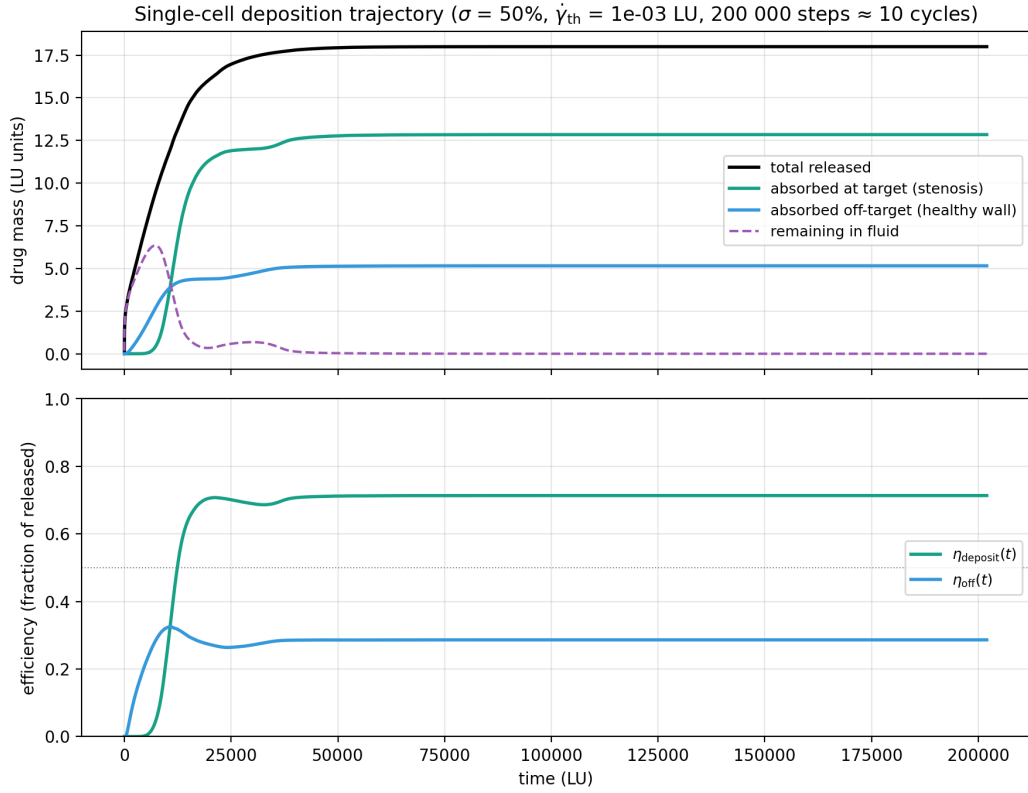


Figure 6. Single-cell deposition cascade for the sweet-spot cell ($\sigma = 50\%$, $\dot{\gamma}_{th} = 1 \times 10^{-3}$ LU, 200,000 production steps). **Top:** cumulative drug mass trajectories. Total released (black) saturates at $M_{loaded} = 18$ by $t \approx 50,000$. Target absorption (green) and off-target absorption (cyan) reach 12.85 and 5.15 respectively, with the residual fluid mass (purple dashed) decaying to near zero. **Bottom:** instantaneous efficiencies $\eta_{deposit}(t)$ and $\eta_{off}(t)$ both reach steady-state plateaus by $t \approx 50,000$, confirming that the 200,000-step run captures the asymptotic deposition value with comfortable margin.

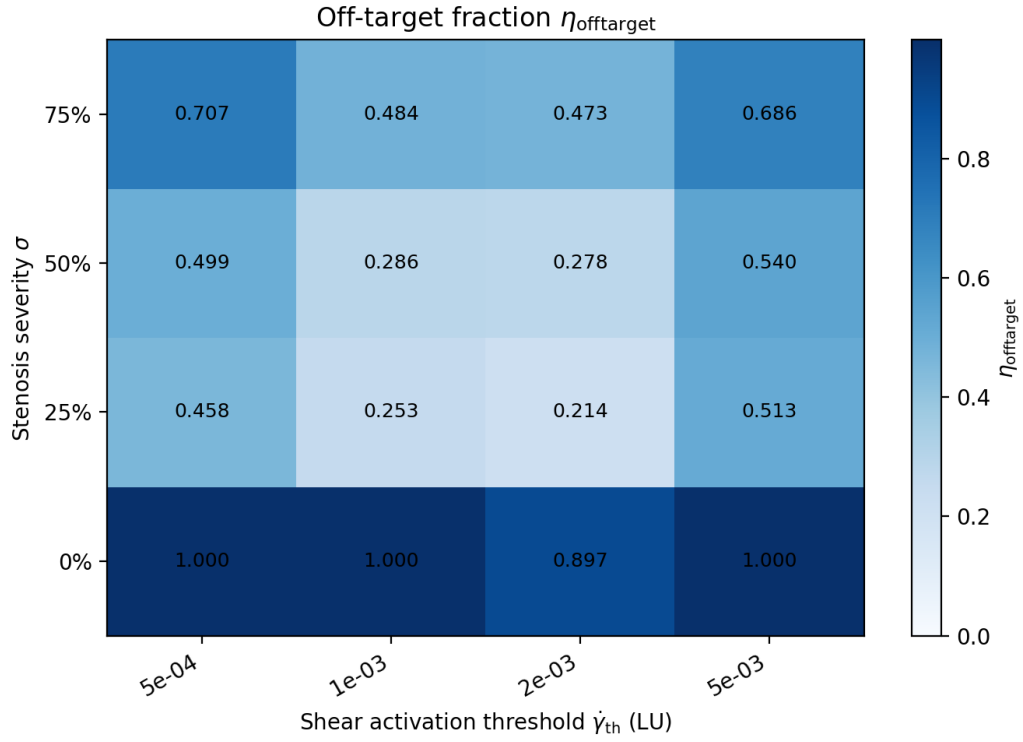


Figure 7. Off-target fraction η_{off} , the fraction of released drug captured at the healthy-wall absorber bands. At the sweet-spot peak ($\sigma = 25\%$, $\dot{\gamma}_{\text{th}} = 2 \times 10^{-3}$) only 21.4% of released drug is lost off-target, giving a $3.6\times$ target:off-target enrichment. The control $\sigma = 0\%$ row gives $\eta_{\text{off}} \rightarrow 1$ (all released drug to healthy wall) by mass conservation.